

Goodarz Firoozi Zargar

Senior Full Stack Engineer | Automation & AI Platforms

Tehran, Iran | +98 937 093 8991 | gf.fantom@gmail.com

linkedin.com/in/goodarzfiroozi | github.com/greygoody

PROFILE

Senior systems-oriented full-stack engineer building automation products, AI-assisted workflows, and developer infrastructure. Turns ambiguous operational work into inspectable software with durable state, explicit review and approval, recoverable execution, and deliberately bounded side effects. Hands-on across TypeScript/Node.js, Python/FastAPI, React/Next.js/SvelteKit, Rust, PostgreSQL, Docker, Linux, and Nix.

CORE TECHNOLOGIES

Backend & Full Stack: TypeScript, Node.js, Python, FastAPI, React, Next.js, SvelteKit, SQL, PostgreSQL, REST APIs

Automation & AI Platforms: MCP, LangGraph, XState, pgvector/RAG, background workers, structured generation, human-in-the-loop workflows

Systems & Delivery: Rust, async and event-driven design, Docker/OCI, Compose, Nix/NixOS, Linux, CI/CD, Git/GitHub, self-hosted infrastructure

PROFESSIONAL EXPERIENCE

Virtual Production & AI Systems Engineer | Rastaar

March 2025 - April 2026 | Tehran / Hybrid

- Served as technical lead for a six-month Unreal Engine virtual-production integration connecting cinematic scene systems, tracked cameras, synchronization, live compositing, distributed computers, and production workflows.
- Built Blueprint-based cinematic-scene systems with granular on-demand level loading and cloud-connected Git/Perforce workflows for distributed art and engineering contributors.
- Integrated and stabilized tracked RED and ARRI ALEXA workflows across VIVE Mars, ReTracker, and Mo-Sys systems, including genlock, calibration, drift control, parallax reduction, and hybrid live/post tracking.
- Contributed to GrandMaster, an AI-assisted game-narrative platform, by prototyping generation, review, backend, and product boundaries that translated creative workflows into explicit software behavior.

Founder & Creative Systems Engineer | Cstudio

February 2019 - Present | Tehran, Iran

- Lead an independent creative-systems and technical R&D practice spanning real-time software, internal tools, interactive systems, and controlled AI-assisted engineering.
- Build and maintain Linux, Docker, Git, VPS, storage, networking, and remote-development infrastructure for distributed creative and engineering work.
- Design proof-oriented Rust, TypeScript, and Python systems with explicit contracts, persisted state, testable boundaries, human approval, and inspectable outputs.

SELECTED ENGINEERING PROJECTS

Multi-Tenant Sports Club Platform

Rust / Axum / Tokio | PostgreSQL | SvelteKit / TypeScript | Docker | Nix | **Hosted demo**

- Designed and delivered an evaluation-ready multi-tenant court-booking and matchmaking platform from an external product brief, combining a tested Rust backend with a hosted TypeScript product surface.
- Modeled tenant-scoped authority, durable PostgreSQL state, booking behavior, and application boundaries; added unit, integration, API, and concurrency tests.
- Built branded member experiences for court availability, booking selection, player matching, fee sharing, events, tenant switching, progress, and separate member/admin views.
- Kept the implementation private while publishing a product demonstration that can be evaluated without transferring source ownership.

SELECTED ENGINEERING PROJECTS, CONTINUED

Energy Ops Intel

Python / FastAPI | Next.js | PostgreSQL / pgvector | LangGraph | Dramatiq / Redis | MinIO | Docker

- Built a private full-stack operational-intelligence platform with a FastAPI backend, Next.js operator workbench, PostgreSQL/pgvector, background workers, object storage, and reproducible local environments.
- Designed a durable AI-assisted contract-review workflow that separates immutable source evidence, typed model proposals, persisted checkpoints, operator decisions, and authoritative application state.
- Added exact evidence-offset validation, interrupt-and-resume review, approve/edit/reject/request-more-evidence decisions, and materialization only after explicit human authorization.
- Maintains a deterministic portfolio demonstration using fictional and sanitized data, without claiming customer deployment, production security, or live-provider reliability.

Shelf Platform

Rust | TypeScript / Node.js | React / Expo | Versioned Commerce Operations | Private Independent Project

- Developing an independently owned commerce-operations platform that separates accepted item facts from inventory state and supports guarded human and AI-assisted changes.
- Built a Rust state and contract layer with expected-version updates, deterministic context packets, typed rejection, concurrency protection, and explicit authority boundaries.
- Added TypeScript operator applications and a runtime boundary for capture, tasks, approvals, events, and adapter-based external integrations, while keeping marketplace-specific implementations non-public.

NixWitness

Rust | Nix/NixOS | Evidence-Guided Verification | github.com/greygoody/NixWitness

- Developing a Rust verification system for deterministic Nix environment inspection and evidence-guided validation of changed projects and system definitions.
- The public repository exposes the reviewed open-source project boundary and public-safe foundation; host-specific proving and unreleased implementation remain private.

EDUCATION

B.Sc. Applied Mathematics | Kharazmi University | 2018

High School Diploma, Mathematics | SAMPAD | 2007 - 2014

PORTFOLIO BOUNDARY

Private project source is available only through a bounded technical review when appropriate. Hosted demonstrations, sanitized walkthroughs, and public-safe documentation are used for initial evaluation.